

MATHCOUNTS®

2024–2025 School Handbook

RESOURCES + INFO EVERY COACH NEEDS

NOW

Look through this handbook and your School Competition Kit.

ASAP

Log in to your **Coach Dashboard** (instructions emailed to you).

AUG. – JAN.

Hold weekly, biweekly or monthly practices with your Mathletes. Use **Coach Dashboard** resources to:

Recruit Students

- How to promote competition math in your school
- How to support competitors of different backgrounds and skill levels

Run Meetings

- What to do during a math team practice
- Suggested calendars
- Tips for making practices fun and easy to plan

Get Math Resources

- Problem of the Week
- Pre-written Practice Plans
- Mini video lessons
- How to use this handbook during team practices

JAN.

Hold the 2025 School Competition (PDF available Nov. 6, 2024 on the **Coach Dashboard**)

JAN. – FEB.

Get ready for the Chapter Competition!

On your **Coach Dashboard**, find:

- Past School + Chapter competitions
- Chapter coordinator contact info
- How to add competitor info for the chapter event
- What to expect on Competition Day

FEB.



Take your Mathletes to the 2025 Chapter Competition!

**LOG IN TO
THE COACH
DASHBOARD**



let's get started!



Get a **FREE** copy of the School Handbook by registering for the
Competition Series at: mathcounts.org/register
or
purchase the complete School Handbook at: [mathcountsstore](https://mathcountsstore.org).

A HANDBOOK PREVIEW

The following three pages provide a preview of the problems in the complete School Handbook.
Answers are at the bottom of this page.

Refer to the last page of this preview for an explanation of all resources (including step-by-step solutions) in the complete School Handbook.

STRETCHES

Each covers a particular math topic.



Prepare students for all rounds: Sprint, Target, Team & Countdown.

WARM-UPS

10 problems
no calculators



Prepare students for the Sprint Round.

WORKOUTS

10 problems
calculators encouraged



Prepare students for the Target Round.

FACTORING STRETCH

1. $2 \times 3^2 \times 5^2 \times 7$
2. 18
3. 72
4. 6210
5. 6976
6. 30
7. 6
8. $11/2$
9. $(1 \pm \sqrt{5})/2$
10. 5

WARM-UP 1

31. 575
32. 64
33. 3
34. 16.3
35. 47
36. 3
37. 562.50
38. 70
39. 14
40. 28

WORKOUT 1

141. 73.50
142. 1
143. 12
144. 9.6
145. 6.98
146. 0.05
147. 550
148. 36
149. 6
150. 87



Factoring Stretch

1. _____ The prime factorization of 315 is $3^2 \times 5 \times 7$. What is the prime factorization of 3150?
2. _____ divisors How many distinct positive divisors does 180 have?
3. _____ divisors If x , y and z are distinct primes, how many distinct positive divisors does $x^3y^5z^2$ have?
4. _____ Tanya finds the least common multiple (LCM) of 180 and 3150. Vanya finds the greatest common divisor (GCD) of 180 and 3150. What is the positive difference between Tanya's number and Vanya's number?
5. _____ What is the sum of the GCD and the LCM of 12^3 and 4^4 ?



FACTORING POLYNOMIALS: In general, the second degree (**quadratic**) polynomial in standard form is $ax^2 + bx + c$. It factors into $a(x - r_1)(x - r_2)$, where r_1 and r_2 are the two roots.

6. _____ When $x^2 - 8x - 209$ is written in factored form, what is the positive difference between the factors?
7. _____ If $x - 4$ is a factor of $x^2 + 13x - 12c + 4$, what is c ?
8. _____ What is the sum of the solutions of $2x^2 - 11x = -12$? Express your answer as a common fraction.

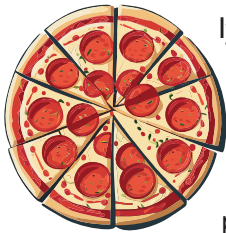
THE QUADRATIC FORMULA: When written in standard form $ax^2 + bx + c$, the two roots of a quadratic polynomial can be found using the **quadratic formula**.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

9. _____ What are the solutions of the equation $x^2 - x - 1 = 0$? Express your answer as a common fraction in simplest radical form.
10. _____ Robin's secret number is $\frac{n-m}{p}$, where $\frac{m \pm \sqrt{n}}{p}$ are the solutions of the equation $x^2 - 7x + 8 = 0$ and m , n and p are relatively prime integers. What is Robin's secret number?



- 
- A wooden treasure chest with a metal latch, filled with gold coins.





Workout 1

141. \$ _____ Pointers Pizzeria offers an 8-inch diameter pizza for \$6. How should they price their large 28-inch diameter pizza, if their goal is to offer the larger pizza at the same price per square inch?

142. _____ What is the 2024th non-zero digit in the decimal expansion of the fraction $\frac{3}{26}$?

143. _____ sets Bucky buys boxes of 12 pencils at a cost of \$3.60 per box. He sells sets of 5 pencils at a cost of \$2.50. How many sets of pencils will Bucky need to sell in order to make a profit of \$12.00?

144. _____ pounds In woodshop class, Addy makes a miniature version of her dining room chair. She creates her mini-chair out of the exact same wood with each dimension being $\frac{1}{4}$ the measurement of her dining room chair. When finished, her mini-chair weighs 2.4 ounces. What is the weight, in pounds, of her dining room chair? Express your answer as a decimal to the nearest tenth. (16 ounces = 1 pound)



145. \$ _____ At a restaurant, Isabella orders three pancakes, each of which costs \$1.80. If she pays 9.25% sales tax on her order, and also gives her server a gratuity of 20% of the original total price of the pancakes (before tax), how much does she pay? Express your answer to the nearest cent.

146. \$ _____ The school cafeteria sells chicken nuggets every Wednesday. You can purchase a half dozen for \$1.80 or 8 for \$2.00. What is the absolute difference in price per nugget between the two options?

147. _____ One-fourth of 56% of N is 77. What is the value of N ?

148. _____ blocks In the game of GIZMOJOJO, wooden blocks each have 4 attributes: color (pink, yellow or green), texture (smooth or bumpy), shape (round or square) and height (tall, short or medium). If there exists exactly one block with each of the different combinations of attributes, how many blocks come in a GIZMOJOJO game?

149. _____ games Prior to her final basketball game, Shelby had scored 88 points. By scoring 23 points in the final game, Shelby raised her season scoring average to 18.5 points per game. How many games did Shelby play during the season?

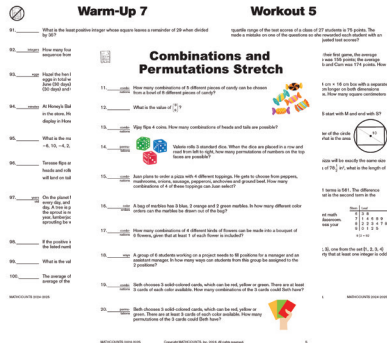


150. _____ What is the sum of the squares of the four smallest prime numbers?

IN THIS YEAR'S HANDBOOK

Problems

**200 MATH PROBLEMS TO
BOOST PROBLEM-SOLVING SKILLS**



11 WARM-UPS

10 questions per Warm-Up
no calculators used



6 WORKOUTS

10 questions per Workout
calculators used

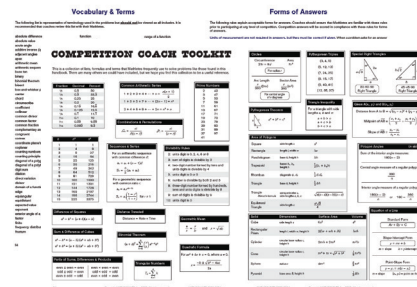


3 STRETCHES

of questions and calculator
use of vary by Stretch

Toolkit

**INCLUDING VOCABULARY
+ FORMS OF ANSWERS**



Answers

INCLUDING DIFFICULTY RATINGS

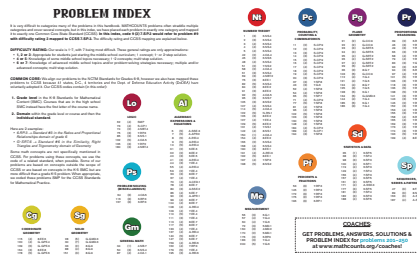
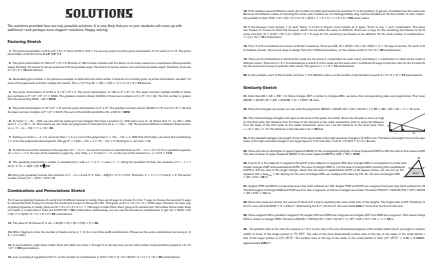
ANSWERS
81. 100
82. 100
83. 100
84. 100
85. 100
86. 100
87. 100
88. 100
89. 100
90. 100
91. 100
92. 100
93. 100
94. 100
95. 100
96. 100
97. 100
98. 100
99. 100
100. 100

Problem Index + CCSS Mapping

ALL 200 PROBLEMS CATEGORIZED

+ MAPPED TO THE CCSS

Solutions
200 STEP-BY-STEP EXPLANATIONS



DON'T MISS OUT! GET YOUR COPY TODAY!